

What Factors Affect International Students' Satisfaction and Migration Intentions?

Course: DSA 210 - Introduction to Data Science

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Document: Final Project Report

1. Motivation

Today, many students travel to different countries for university education. For universities, these international students are very important because they bring different cultures, talent, and financial support. However, moving to a new country and starting a new education system can be very difficult for students. We need to understand what makes these students happy or unavailable. If universities know these factors, they can help students feel better and give them a better education. Also, understanding student satisfaction helps us see their future plans and migration choices (whether they want to stay in that country after graduation or not). In this project, we want to use data science methods to see how student details, scholarships, tuition costs, and school quality affect the overall satisfaction of international students.

2. Data Source

For this project, I used a dataset named `world_university_survey_dataset.csv`, which I found on Kaggle. It contains survey answers from 1,000 different international students. The data has 15 columns in total. These columns include personal information (like age, gender, country), academic details (university, program level, field of study), financial information (tuition fee in USD, scholarship status), and school ratings (campus facilities rating, teaching quality rating). The most important column (our target variable) is `overall_satisfaction`. In the original data, satisfaction was written as text (like "Very Dissatisfied", "Neutral", "Satisfied"). To do statistical tests and build machine learning models, I converted these text levels into numbers from 1 to 5 and created a new column named `satisfaction_score`.

3. Data Analysis

I analyzed the data in two main steps: Statistical Hypothesis Testing and Machine Learning Modeling.

Statistical Hypothesis Testing:

- **Independent t-test (Welch's t-test):** I used this test to compare the average satisfaction score between students who have scholarships and students who do not. This helps us see if financial help makes students happier.
- **One-Way ANOVA Test:** I used this test to check if satisfaction scores change depending on the student's program level (Bachelor, Master, or PhD). This shows us if older or more advanced students have different satisfaction levels.
- **Chi-Square Test of Independence:** I used this test to see if there is a relationship between taking online classes and overall satisfaction. This helps us understand if the way classes are taught changes student happiness.

Machine Learning Modeling:

- I defined this problem as a multi-class classification task. The goal is to predict the exact category of `overall_satisfaction`.
- **Data Cleaning & Preprocessing:** First, I dropped the `satisfaction_score` column to prevent data leakage because it was identical to our target variable. Then, I used Label Encoding to change categorical columns (like country and gender) into numbers. I also used `StandardScaler` to normalize numerical columns so they are all on the same scale. Finally, I split the data into 80% for training the models and 20% for testing them.
- **Models Used:** I used two different algorithms. First, I trained a **Logistic Regression** model as a simple baseline. Second, I trained a **Random Forest Classifier** (with 100 trees) because it can find complex, non-linear relationships in survey data better than simple models.

4. Findings

Our hypothesis tests gave us important insights. The Welch's t-test and ANOVA tests showed us how satisfaction changes when students receive financial support or move to higher degree levels. The Chi-Square test helped us see if online classes have a strong relationship with student satisfaction levels.

When I compared the two models, the Random Forest Classifier performed better than the baseline Logistic Regression. The Random Forest model used campus scores (teaching quality rating, campus facilities rating) and financial numbers (tuition fee in USD) effectively to make correct predictions. I checked the success of the models by looking at Accuracy, Precision, Recall, and F1-score values in the classification report.

5. Limitations and Future Work

Limitations: The main limitation is that this dataset comes from a static survey. It does not show real-time economic changes or visa policy changes in host countries, which can affect student plans. Also, "satisfaction" is a very personal and subjective feeling, so simple number scales from 1 to 5 might not show the full truth about a student's experience.

Future Work: In the future, I can make this project better by adding external data, such as the living costs (Cost of Living Index) in different countries, job opportunities for international graduates, and visa rules for different nationalities. Also, I can try more advanced algorithms like XGBoost and use hyperparameter tuning (GridSearchCV) to get higher prediction accuracy.

AI Disclosure (Academic Integrity)

In accordance with the academic honesty policy of this course, I declare that I completed this project entirely on my own, including data collection, statistical analysis, and machine learning modeling. An AI tool was used very minimally, only at the final stage for basic proofreading and document formatting.

- **Prompts used:** I provided my own completed analysis points and text drafts to check for grammatical errors and fix paragraph formatting.
- **Outputs generated:** The tool assisted in correcting minor English grammar mistakes and organizing my original text into the required professional report structure.